

Milestone 2 Progress Report

Project Title: MedAssist AI – Medical Symptom Analysis & Disease Prediction System

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1. Objective

The objective of Milestone 2 was to transform the initial machine learning prototype into a complete backend healthcare platform. The work focused on integrating the ML model with FastAPI, PostgreSQL database, secure authentication, report generation, analytics, profile management, medical history, and administrator functionalities.

2. Work Completed

Database Integration

Designed relational database using PostgreSQL with Users, Admins, Predictions, Reports and Medical History tables. Implemented SQLAlchemy ORM models and established relationships.

Authentication

Implemented patient registration, patient login, admin login, JWT authentication and BCrypt password hashing.

Disease Prediction

Integrated the trained Random Forest model. Symptoms are converted into feature vectors and disease predictions are generated with confidence scores.

Risk Assessment

Implemented risk score calculation and categorized results into Low, Medium, High and Critical.

Recommendation

Integrated disease description, precautions, diet, lifestyle and medicine guidance from healthcare datasets.

Report Generation

Generated professional PDF reports containing patient details, prediction, confidence, risk level and recommendations.

Medical History

Automatically stores every completed prediction with report references for future access.

Profile

Implemented profile update functionality including phone, address, emergency contact and profile image.

Patient Analytics

Implemented APIs for disease distribution, risk distribution and monthly prediction trend.

Admin Module

Completed View All Patients, View Single Patient, Search Patient, View Reports, High Risk Patients and Dashboard Statistics.

Admin Analytics

Implemented Disease Distribution, Risk Distribution, Monthly Prediction Trend, Most Common Diseases and Most Common Symptoms.

3. Technologies Used

Backend: FastAPI, Python

Database: PostgreSQL, SQLAlchemy ORM

Machine Learning: Scikit-learn, Pandas, NumPy, Random Forest

Security: JWT, BCrypt

Reports: ReportLab

Testing: Swagger UI

Version Control: Git & GitHub

4. Current Module Status

Module	Status
Database	Completed
Authentication	Completed
Disease Prediction	Completed
Risk Assessment	Completed
Recommendation	Completed
Health Reports	Completed
Medical History	Completed
Profile	Completed
Patient Analytics	Completed
Admin	Completed
Admin Analytics	Completed
API Testing	Completed

5. Challenges Faced

Major challenges included integrating machine learning predictions with REST APIs, designing a normalized database, implementing secure authentication, maintaining relationships between reports and prediction history, and developing reusable analytics APIs. These challenges were resolved through modular backend design and extensive Swagger testing.

6. Next Milestone

The next phase will focus on React.js frontend development, Bootstrap UI implementation, API integration, patient dashboard, admin dashboard, complete end-to-end testing, deployment and the

Forgot Password enhancement.

7. Conclusion

Milestone 2 successfully established the complete backend foundation of MedAssist AI. The backend now supports authentication, disease prediction, risk assessment, recommendation generation, PDF report creation, medical history, profile management, patient analytics and administrator operations. With backend development completed, the project is ready for frontend integration.